

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

BOARD ORDER NO. 6-99-37
WDID NO. 6B360115002

REVISED WASTE DISCHARGE REQUIREMENTS

FOR

SAN BERNARDINO COUNTY DEPARTMENT OF AIRPORTS;
BARSTOW-DAGGETT AIRPORT WASTEWATER TREATMENT FACILITY

San Bernardino County

The California Regional Water Quality Control Board, Lahontan Region (Regional Board) finds:

1. Discharger

San Bernardino County Department of Airports submitted a complete Report of Waste Discharge for the Barstow-Daggett Airport Wastewater Treatment Facility on June 15, 1998. For the purposes of this Regional Board Order (Order), San Bernardino County Department of Airports is referred to as the "Discharger."

2. Facility

The Barstow-Daggett Airport Wastewater Treatment Facility is the facility from which the discharge occurs. For the purposes of this Order, the Barstow-Daggett Airport Wastewater Treatment Facility is referred to as the "Facility." The upgraded Facility will discharge municipal wastewater.

3. History of Previous Regulation by the Regional Board

The Regional Board previously revised Waste Discharge Requirements (WDRs) for the Facility under Board Order No. 6-94-58, which was adopted on May 19, 1994.

4. Reason for Action

The Regional Board is revising the WDRs to reflect the Discharger's proposed renovation and upgrade in the Facility's treatment processes to provide a secondary-level of treatment and relocation of the percolation disposal ponds.

5. Facility Location

The Facility is located approximately six miles east of the Community of Daggett, San Bernardino County, near the northwest corner of the Barstow-Daggett Airport (Airport) complex, Section 21, T9N, R2E, SBB&M, as shown on Attachments "A" and "B," which are made part of this Order.

6. Description of Facility and Discharge

The San Bernardino County Department of Airports operates a wastewater collection, treatment and disposal Facility for the Barstow-Daggett Airport. The Airport includes a U.S. Army Heliport and other commercial tenants. An estimated average of 0.015 million gallons per day (mgd) of raw sewage is collected from residential and commercial facilities on the Airport property and currently conveyed to an existing holding tank with final disposal to two unlined facultative wastewater oxidation ponds for biological treatment and final disposal by evaporation and percolation. The proposed upgrade of the Facility consists of an influent pump station, two comminuting grinders, one magnetic flowmeter, two lined aerated lagoons connected in parallel, two lined aerated polishing ponds connected in parallel, two unlined evaporation/percolation ponds connected in parallel, and three ground water monitoring wells. The two aeration ponds, to cover a total of 0.3 acres, are proposed to be lined with six inches of concrete, and each fitted with two floating aerators. The two aerated polishing ponds to cover a total of 0.2 acres are proposed to be lined with six inches of concrete, and each fitted with one floating aerator. The two evaporation/percolation ponds are proposed to cover a total of 0.4 acres. The system is designed for an average daily design flow of 0.15 mgd. The maximum hydraulic capacity is 0.45 mgd. The Airport population consists of approximately 50 permanent residents (military personnel and airport employees), public patronage and military day-use personnel (estimated at 250 people per day, on average; a portion of the military component of these 250 day-use personnel may be relocated to residence at the Airport).

The effluent is existing municipal wastewater from the Airport, with a proposed intermittent minor component of pretreated helicopter washrack and hangar floor-drain wastewater. Washrack, and any floor-drain wastewater are proposed to be treated first in an oil/water separator and then in an on-site package wastewater pretreatment plant to provide chemical coagulation, flocculation, sedimentation, filtration and disinfection prior to recycling the treated water for washrack reuse. The package washrack wastewater pretreatment plant is proposed to intermittently discharge a low-flow wastewater component, which would be commingled with the domestic wastewater influent to the upgraded Facility.

7. Sludge Treatment and Disposal

The Facility pond treatment and disposal system will generate sludge which is projected to require pond cleaning approximately once every four years. Each pond unit process has two ponds which readily allows one pond in each pair of ponds to be taken offline for maintenance and cleaning operations. Removed sludge will be transported offsite for disposal in an authorized class III landfill disposal site.

8. Disposal Area

The designed hydraulic disposal capacity of the two-pond evaporation/percolation disposal system is 0.15 mgd. The ponds can be alternately operated to allow for maintenance.

9. Authorized Disposal Site

The two-pond 0.4-acre evaporation/percolation pond system is the only authorized disposal site for wastewater. The authorized disposal site is located on land owned and administered by the County of San Bernardino.

10. Site Geology

The Facility is located in a portion of the valley formed by the Mojave River, at an elevation of approximately 1,920 feet (mean sea level), on flat to very mildly sloped terrain underlain by unconsolidated gravel, sand, and silt. The nature of the area is semi-arid high desert, with scattered rock outcrops and mountainous areas to the north and south. Predominant vegetation consists of succulents and sage scrub.

11. Site Hydrology

The nearest surface water body is the (ephemeral) Mojave River, located approximately two miles north of the Facility. Due to the flat terrain in the vicinity of the Facility, potential for surface runoff is minimal. Seasonal high temperatures may average over 100°F in the summer months and contribute to high evaporation rates.

12. Site Hydrogeology

Two wells supply drinking water to the Airport; these wells are located approximately 3,000 feet southwest of the oxidation pond area. Depth to ground water in these wells in early 1993 was 116 to 118 feet below the ground surface. The water supply system is also being upgraded by addition of a new well and storage tanks as a part of the Airport renovation and heliport project. Depth to ground water beneath the Facility has been estimated at 80 feet or more, and is assumed to flow from south to north, following the topographic gradient, towards the Mojave River. The quality of the ground water is excellent for all beneficial uses. Ground water monitoring is required for the upgraded Facility.

13. Receiving Waters

The receiving waters are the ground waters of the Lower Mojave River Valley Ground Water Basin (Department of Water Resources Ground Water Basin No. 6-40).

14. Lahontan Basin Plan

The Regional Board adopted a Water Quality Control Plan which became effective on March 31, 1995 for the Lahontan Region (Basin Plan) and this Order implements the Basin Plan as amended.

15. Beneficial Uses

The beneficial uses of the ground waters of the Lower Mojave River Valley Ground Water Basin as set forth and defined in the Basin Plan are:

- a. municipal and domestic supply;
- b. agricultural supply;
- c. industrial service supply; and
- d. freshwater replenishment.

16. California Environmental Quality Act Compliance

The County of San Bernardino is engaged in the continued operation of an existing Facility, and, as such, is exempt from the requirements of the California Environmental Quality Act (Public Resources Code, § 21000, et seq.).

17. Notification of Interested Parties

The Regional Board has notified the Discharger and interested parties of its intent to revise WDRs for the discharge.

18. Consideration of Public Comments

The Regional Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that the Discharger shall comply with the following:

I. DISCHARGE SPECIFICATIONS

A. Effluent/Discharge Limitations

1. The total effluent flow of wastewater during a 24-hour period shall not exceed 0.15 million gallons.
2. All wastewater made available for percolation at the authorized disposal sites shall not contain concentrations of parameters in excess of the following limits:

<u>Parameter</u>	<u>Units</u>	<u>Mean¹</u>	<u>Maximum</u>
Biochemical Oxygen Demand ²	mg/l	30	45
Methylene Blue Active	mg/l	1.0	2.0

All wastewater made available for percolation at the authorized disposal sites shall have a pH of not less than 6.0 pH units nor more than 9.0 pH units.

All wastewater made available for percolation at the authorized disposal sites shall have a dissolved oxygen concentration not less than 1.0 mg/l.

¹ The arithmetic mean of lab results for effluent samples collected in a period of 30 consecutive days.

² Biochemical oxygen Demand (5 day, 20°C) of a filtered sample using a No. 1 Whatman filter or equivalent and reseeded with an unfiltered sample.

B. Receiving Water Limitations

1. The discharge of waste shall not cause the presence of the following substances or conditions in ground waters of the South Lahontan Hydrologic Basin, Mojave Hydrologic Unit, Lower Mojave Hydrologic Area:
2. Any perceptible color, odor, taste or foaming.
3. Coliform organisms attributable to human wastes.
4. Toxic substances in concentrations that individually, collectively, or cumulatively cause detrimental physiological responses in human, plants, animals, or aquatic life.
5. Identifiable chlorinated hydrocarbons, organophosphates, carbamates, and other pesticide and herbicide groups, in summations, in excess of the lowest detectable levels.
6. Concentrations of chemical constituents in excess of the maximum contaminant levels or secondary maximum contaminant levels based upon drinking water standards specified by the more restrictive of the California Code of Regulations, Title 22, Division 4, Chapter 15, or 40 CFR, Part 141.

C. General Requirements and Prohibitions

1. There shall be no discharge, bypass, or diversion of raw or partially treated sewage, sewage sludge, grease, or oils from the collection, transport, treatment, or disposal facilities to adjacent land areas or surface waters.
2. Surface flow or visible discharge of sewage or sewage effluent at, or from, the authorized disposal site to adjacent land areas or surface waters is prohibited.
3. The vertical distance between the liquid surface elevation and the lowest point of a pond dike or the invert of an overflow structure shall not be less than two (2) feet.
4. The discharge shall not cause a pollution as defined in Section 13050 of the California Water Code, or a threatened pollution.
5. Neither the treatment nor the discharge shall cause a nuisance as defined in Section 13050 of the California Water Code.
6. The discharge of wastewater except to the authorized disposal site is prohibited.

II. PROVISIONS

A. Rescission of Waste Discharge Requirements

Board Order No. 6-94-58 shall remain in affect for the existing Facility and is hereby rescinded upon initiation of discharge to the upgraded Facility.

B. Standard Provisions

The Discharger shall comply with the "Standard Provisions for Waste Discharge Requirements," dated September 1, 1994, in Attachment "C", which is made part of this Order.

C. Monitoring and Reporting

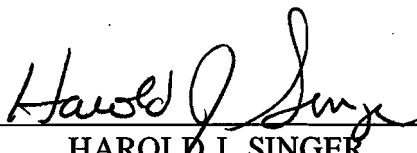
1. Pursuant to the California Water Code Section 13267(b), the Discharger shall comply with Monitoring and Reporting Program No. 94-58 for the existing Facility as specified by the Executive Officer until discharge commences to the upgraded Facility. Thereafter, the Discharger shall comply with Monitoring and Reporting Program No. 99-37 as specified by the Executive Officer.
2. The Discharger shall comply with the "General Provisions for Monitoring and Reporting," dated September 1, 1994, which is attached to and made part of the Monitoring and Reporting Program.

D. Time Schedule Compliance

1. At least 30 days prior to initiation of discharge from the upgraded Facility to the new percolation ponds, the Discharger shall submit a completion report signed by a Registered Civil Engineer, California Certified Engineering Geologist or a California Registered Geologist documenting the following:
 - a. the installation of a minimum of three monitoring wells to determine the gradient of ground water flow; of which two wells are downgradient of the new percolation ponds;
 - b. the placement, lithology and construction of the wells;
 - c. the well design and construction conforms to California Well Standards, Department of Water Resources Bulletins 74-90 and 74-81; and
 - d. that the well permits were obtained from the San Bernardino County Department of Environmental Health Services. Prior to installation, the conceptual design and placement of the wells shall be submitted.

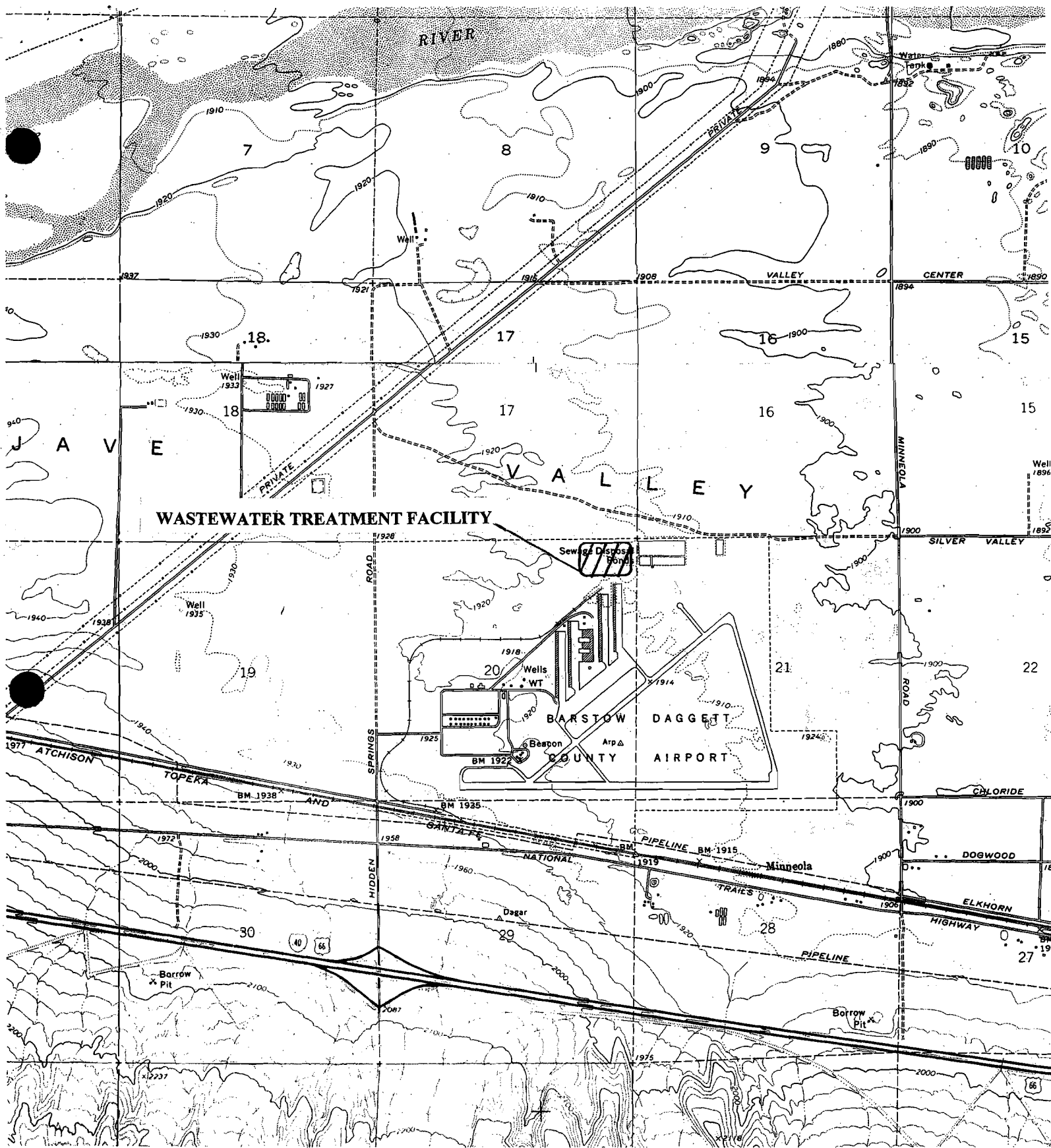
2. Within 30 days after completion of construction of the upgraded Facility, the Discharger shall submit a report signed by a Registered Civil Engineer, a California Certified Engineering Geologist or a California Registered Geologist that the Facility was designed and constructed as proposed.
3. By **March 31, 2000**, the Discharger shall submit a closure report including a plan and time schedule with completion date for closure of the old wastewater holding tank, pump station, and old percolation ponds. The plan shall specify how these facilities shall be closed in a manner protective of health, safety and water quality.

I, Harold J. Singer, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Lahontan Region, on September 9, 1999.



HAROLD J. SINGER
EXECUTIVE OFFICER

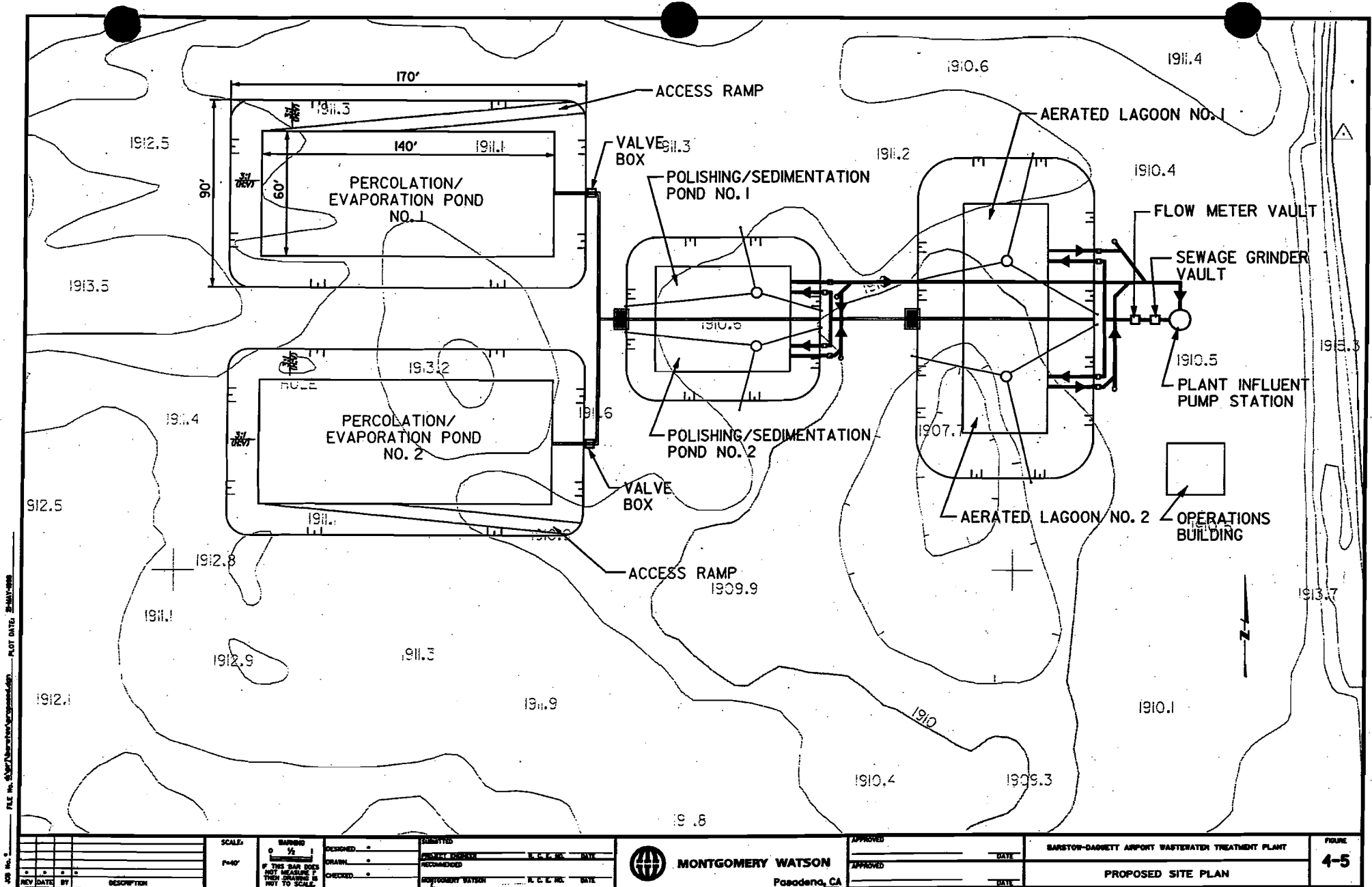
Attachments: A. Location Map
B. Wastewater Facility Layout
C. Standard Provisions for Waste Discharge Requirements



ATTACHMENT "A"

SAN BERNARDINO COUNTY DEPARTMENT OF AIRPORTS BARSTOW-DAGGETT AIRPORT WASTEWATER TREATMENT FACILITY

WITHIN THE NE/4, SECTION 20, T9N, R2E, SBB&M
USGS MINNEOLA 1971 7.5 MINUTE QUADRANGLE



WASTEWATER TREATMENT FACILITY LAYOUT

ATTACHMENT "B"

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LAHONTAN REGION

MONITORING AND REPORTING PROGRAM NO. 99-37
WDID NO. 6B60115002

FOR

SAN BERNARDINO COUNTY DEPARTMENT OF AIRPORTS;
BARSTOW-DAGGETT AIRPORT WASTEWATER TREATMENT FACILITY

San Bernardino County

I. MONITORING

A. Flow Monitoring

The Discharger shall monitor the following:

1. The total volume, in million gallons, of wastewater flow to the Facility for each day.
2. The total volume, in million gallons, of wastewater flow to the Facility for each month.
3. The average flow rate, in million gallons per day (mgd), of wastewater to the Facility calculated for each month.
4. The maximum instantaneous flow rate, in mgd, of wastewater to the Facility that occurs each day.
5. The freeboard (vertical distance from the top of the lowest part of the dike to the wastewater surface in the pond) measured each month in each surface impoundment. If a surface impoundment does not contain wastewater, indicate that it is empty.

B. Facility Effluent Monitoring

Beginning immediately, eight-hour composite samples of the effluent from the Facility shall be collected and analyzed to determine the magnitude of the following parameters:

<u>Parameter</u>	<u>Units</u>	<u>Frequency</u>
Biochemical Oxygen Demand	mg/l	Monthly
Chemical Oxygen Demand	mg/l	Monthly
Methylene Blue Active Substances	mg/l	Monthly
pH	pH units	Monthly

Beginning immediately, grab samples of the effluent shall be collected and analyzed to determine the magnitude of the following parameters:

<u>Parameter</u>	<u>Units</u>	<u>Frequency</u>
Oil & Grease	mg/l	Quarterly
Total Petroleum Hydrocarbons	mg/l	Quarterly
Purgeable Organics ¹	ug/l	Quarterly
Base/Neutral/Acid Extractables ¹	ug/l	Quarterly
Kjeldahl Nitrogen	mg/l as N	Quarterly
Nitrate Nitrogen	mg/l as N	Quarterly
Ammonia Nitrogen	mg/l as N	Quarterly
Dissolved Oxygen	mg/l	Quarterly
Heavy Metals ¹	mg/l	Quarterly
Total Dissolved Solids	mg/l	Quarterly
Chloride	mg/l	Semiannually
Sodium	mg/l	Semiannually
Sulfate	mg/l	Semiannually
Total Hardness as CaCO ₃	mg/l	Semiannually

C. Ground Water Monitoring

1. By 30 days prior to initiation of discharge to the new percolation ponds, the Discharger shall submit a ground water monitoring system plan and a time schedule for installation of ground water monitoring wells described below. The monitoring system, at a minimum shall, include the following:
 - a. Three proposed wells shall be installed, with at least one (1) upgradient and two (2) downgradient of the upgraded wastewater treatment Facility's new evaporation percolation disposal ponds and leachline. Use of an existing background water quality monitoring well may be proposed if the location is appropriate. The Discharger shall demonstrate that at least two downgradient wells are located such that ground water potentially impacted by the discharge will be monitored.
 - b. The specific design and location of the wells shall be submitted for approval.

¹ Analyze for all appropriate substances included on the US EPA list of Priority Pollutants (attached) and all other toxic substances known to be discharged to the Discharger's sewer system. The list of Priority Pollutants is found in 40 CFR 131.36 (b).

- c. An as-built design report shall be submitted within 60 days after the ground water monitoring system is installed. This shall include a statement of certification signed by a California Registered Civil Engineer, or Geologist regarding the placement, lithology and construction of the wells.
 - d. Prior to the initiation of discharge to the new evaporation/percolation ponds, the three new additional ground water monitoring wells shall be installed in accordance with the submitted plan and a baseline sample collected and analyzed for the constituents listed below in Ground Water Sampling requirement No. C.2.
2. In accordance with the noted frequencies, grab samples from the entire thickness or the upper 20 feet, whichever is less, of the uppermost ground water bearing zone shall be collected from the monitoring wells and analyzed to determine the magnitude of the following parameters:

<u>Parameter</u>	<u>Units</u>	<u>Frequency</u> ²
Methylene Blue Active Substances	mg/l	Quarterly
Chloride	mg/l	Quarterly
Total Dissolved Solids	mg/l	Quarterly
Kjeldahl Nitrogen	mg/l	Quarterly
Nitrate Nitrogen	mg/l as N	Quarterly
Purgeable Organics ¹	ug/l	Annually
Base/Neutral/Acid Extractables ¹	ug/l	Annually

- a. The depth to ground water in each well shall be measured and recorded each time a monitoring well is sampled.
- b. The velocity and direction of ground water flow under the Facility site shall be determined at least semi-annually unless it can be shown that no changes have occurred.
- c. Methylene blue active substances, chloride, total dissolved solids, and nitrate nitrogen are to be sampled and analyzed quarterly for the first year, and annually thereafter.

² Methylene Blue Active Substances, Chloride, Total Dissolved Solids, Kjeldahl Nitrogen, Ammonia Nitrogen and Nitrate Nitrogen are to be sampled and analyzed quarterly for the first year, and semi-annually thereafter.

- d. Well Purging: Ground water samples shall be taken only after at least three volumes of water in the casing have been removed and/or temperature, electrical conductivity, and pH of water in the well have stabilized. Low flow purging is preferred.
- e. Total Dissolved Solids (TDS) concentrations in the background (upgradient) monitoring well and the downgradient monitoring wells shall be compared during each ground water sampling event. The change in TDS concentration between the upgradient well and each downgradient well shall be determined.

D. Sludge Monitoring

- 1. The Discharger shall submit an updated Sludge Management Plan by March 31, 2000.
- 2. The Discharger shall provide any deviations from the Sludge Management Plan.
- 3. Total quantity of sludge generated (cleaned from any ponds) during the monitoring period.
- 4. Date and quantity of any sludge landfilled or removed off site, location of use, recipient (including name and address) and sludge disposal method (including crops grown, if appropriate) for all sludge moved offsite.
- 5. Cumulative total quantity of sludge currently stockpiled on site, including the quantity of sludge added to the stockpile during this monitoring period.
- 6. For sewage sludge removed from ponds, stockpiled or stored onsite, or discharged offsite during the previous month, a representative sample shall be collected and analyzed for the following constituents:
 - a. total Kjeldahl nitrogen as N;
 - b. ammonia nitrogen as N;
 - c. nitrate nitrogen as N;
 - d. total phosphate as P; and
 - e. parameters listed in Title 22, Division 4.5, Chapter 11, Article 3, §66261.24, (2) (A) and (2XB) of the California Code of Regulations.

7. The Discharger shall make a determination whether the analyses indicate that the sludge shall be considered a hazardous material.
8. At least 90 days prior to the first sludge sample collection and analysis, a sludge sampling protocol shall be submitted to the Executive Officer for approval.

II. REPORTING

A. General Provisions

The Discharger shall comply with the "General Provisions for Monitoring and Reporting," dated September 1, 1994, which is attached to and made part of this Monitoring and Reporting Program.

B. Submittal Periods

Beginning on the 15th day of the month following the adoption of this Order, a monitoring report including the preceding information shall be submitted to the Regional Board. Subsequent monitoring reports shall be submitted to the Regional Board by the 15th day of the month following each quarter.

C. Time Schedules

The Discharger shall submit a Sampling and Analysis Plan in accordance with General Provisions for Monitoring and Reporting Program No. 1.d. for all sampling required at the Facility (soil, ground water, effluent, sludge, etc.).

Ordered by: Harold J. Singer Dated: September 9, 1999
HAROLD J. SINGER
EXECUTIVE OFFICER

Attachments: A. General Provisions for Monitoring and Reporting
B. Priority Pollutants